

CE 565P: Experimental Techniques in Hydrology

Course Code: CE 565P

Course Name: Experimental Techniques in Hydrology

L-T-P-C: 0-0-4-2

Prerequisites: NA

Intended for: UG/PG Elective

Mutual Exclusion: NA

Approval: 60th BoA

Preamble

This course aims to provide a comprehensive understanding of the methods and practices used in experimental hydrology. As the name suggests, the focus of the course will be on applying hydrological theories through practical experiments and measurement of hydrological characteristics.

In this course, students will have the opportunity to engage with a range of lab-based activities to understand the physical properties of soils, plot soil moisture retention curves, estimate infiltration rates, simulate rainfall scenarios, measure soil hydraulic conductivity, and study the flow dynamics in open channels. These activities will not only help students in grasping the underlying scientific principles but also in developing the skills required to design and conduct hydrological experiments.

The hands-on nature of the course will allow the students to translate theory into practice, fostering a greater appreciation for techniques in experimental hydrology.

Laboratory/Practical/Tutorial Modules

- **Module 1 - Determination of Physical Properties of Soil:** Theoretical and practical understanding of techniques to determine soil physical properties. (4 hours)
- **Module 2 – Estimation of Soil Moisture Retention Characteristic Curve:** Theory and application of pressure plate apparatus to plot soil moisture retention curve. (4 hours)
- **Module 3 – Infiltration Capacity of Soil:** Estimation of soil infiltration capacity (in lab and field) using various techniques. (6 hours)
- **Module 4 - Soil Hydraulic Conductivity Experiments:** Hands-on exercises on understanding and measuring soil hydraulic conductivity. (4 hours)
- **Module 5 - Rainfall Simulator Experiments:** Simulating various rainfall scenarios by varying the catchment characteristics and understanding their impacts on runoff generation for various soil types. (10 hours)
- **Module 6 - Flow through Open Channel Experiments:** Experiments to understand and analyze flow dynamics in open channels and their interaction with subsurface. (10 hours)
- **Module 7 – Hydrometry:** Methods of in-situ measurement of critical hydrological elements i.e. - rainfall, soil moisture, and streamflow. (8 hours)
- **Module 8 – Geophysical investigation using non-intrusive techniques:** To understand and use the non-intrusive method to map underground utilities and subsurface characteristics. (6 hours)
- **Module 9: Field Visit:** A half-day field visit to nearby hydrometeorological sites. (4 hours)



Text Books

1. Brassington, R. (2023). *Field hydrogeology*. John Wiley & Sons.

References

1. WMO's *Guide to Hydrological Practices*, Volume I & II. https://library.wmo.int/index.php?lvl=notice_display&id=21815
2. IMD's *Instruction to Observers at the Surface Observatories*. https://www.imdpune.gov.in/library/instructions_to_observers.pdf
3. CWC's *Handbook for Hydrometeorological Observations*. https://cwc.gov.in/sites/default/files/hand-book-hydro-meteorological-observations._1.pdf